

CURRICULUM VITAE

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SUMMARY

A highly motivated **Robotics and Ocean Engineer** with a Master's thesis and research experience on **collision avoidance for autonomous ships**. I have published a journal article on collision avoidance for ASVs using Artificial Potential Fields. I possess extensive hands-on experience in implementing and validating control and guidance algorithms on scaled marine vessels using ROS, Python, and MATLAB. My core competencies include control theory, motion planning (APFs), dynamical systems, and machine learning.

PROFESSIONAL PREPARATION

ZF Group, India

Senior Engineer, Vehicle Motion Control

Hyderabad, India

July 2023 - Current

Indian Institute of Technology Madras

- B.Tech. (Naval Architecture and Ocean Eng.)
- M.Tech. (Robotics) (Combined Dual Degree CGPA: 8.65/10)

Chennai, India

July 2018 - July 2023

RESEARCH ARTICLES

Journal Publications

- **Jadhav, A. K.**, Pandi, A. R., & Somayajula, A. (2023). "Collision avoidance for autonomous surface vessels using novel artificial potential fields." *Ocean Engineering*, 288(Part 2), 116011. <https://doi.org/10.1016/j.oceaneng.2023.116011>

Conference Proceedings

- Deogaonkar, VV, **Jadhav, AK**, Ramachandran, K, & Somayajula, AS. "Data Driven Identification of Ship Maneuvering Coefficients." *Proceedings of the ASME 2023 42nd International Conference on Ocean, Offshore and Arctic Engineering. Volume 5: Ocean Engineering*. Melbourne, Australia. June 11–16, 2023. V005T06A050. ASME. <https://doi.org/10.1115/OMAE2023-104644>

RESEARCH EXPERIENCE

MAV* Lab, Center for Marine Autonomous Systems, IIT Madras

(May 2021 - July 2023)

Development of Autonomy and Experimental Validation on Institute Lake ([Video Link](#))

- **Spearheaded the design and execution of over 15 field experiments** with a 1:75.5 scaled autonomous vessel, successfully validating the real-world performance of autonomous guidance and control strategies thus directly addressing the challenge of implementing research on physical marine platforms
- **Developed a data-driven approach to estimate ship maneuvering** by applying system identification techniques to real-world trial data, enabling a robust foundation for simulation and the design of control systems
- **Implemented and fine-tuned a low-level control stack**, featuring a PD heading controller and an Integral Line-of-Sight (I-LoS) guidance system for precise autonomous waypoint tracking, demonstrating core competency in marine vehicle control
- **Assisted in creating a modular autonomy pipeline** using Docker container to integrate and deploy a Reinforcement Learning (RL) agent a scaled model ship, enabling real-time, adaptive policy execution based on environmental feedback

Design and Fabrication of a 1:75.5 Scale Kriso Container Ship (KCS)

- Conducted detailed resistance analysis and hull calculations using theoretical models to estimate hydrodynamic performance at 0.26 Froude Number (1.42 m/s).
- Derived propulsion system specifications from resistance estimates
- Applied CAD tools (Fusion 360) for 3D Modeling of the propulsion system.

PROFESSIONAL EXPERIENCE

ZF Group, India (*Senior Engineer, Vehicle Motion Control*)

(July 2023 - Present)

Project 1 - Electromechanical Roll Control for Active Roll Stabilizers

- Implemented and validated a Lyapunov stable control logic to offer better roll comfort of a vehicle for below 4Hz road disturbance frequencies.
- **Analyzed in-vehicle CAN signal data** from experimental trials to evaluate ride comfort and control tracking performance. Used asammdf python library to extract and analyze the vehicle experimental data in MATLAB.
- Extracted roll acceleration characteristics and assessed controller behavior using frequency-domain techniques including **FFT** and **Bode plot analysis**.
- Created and validated vehicle roll acceleration module in simulation to evaluate qualitative comfort gain using different control strategies.

Project 2 - Adaptive AI-based Feedforward Control for Lateral Vehicle Dynamics

- Developed an end-to-end pipeline emulating a closed-loop data system for **Software Defined Vehicles**, enabling continuous improvement of an AI-based lateral control module in a vehicle.
- This pipeline created a framework for the vehicle to **fine-tune internal models based on unseen data**, demonstrating an application of data-driven controller refinement.
- Implemented a MATLAB-Simulink simulation of vehicle dynamics and open-loop control to generate training data based on road curvature and vehicle velocity profiles.
- Integrated Python-based Multilayer Perceptron (MLP) training scripts using Azure ML services in MATLAB to iteratively fine-tune the neural network on newly observed data.
- Designed inference and performance analysis routines within MATLAB to evaluate controller tracking accuracy via simulation and ensure robustness of weight updates before deployment.
- Demonstrated **Over-the-Air (OTA)** update workflows by automating replacement of model weights in the simulation loop, effectively mimicking real-world cloud-to-vehicle integration.

COURSEWORK AND SKILLS

Academic Curriculum IITM and other Coursework

- Principles of Guidance for AVs
- Control of Automotive Systems
- Field and Service Robotics
- Mechanics and Control of Serial Robots
- Introduction to Robotics
- Udacity Deep Learning Nanodegree
- Ship Dynamic Positioning System
- Reinforcement Learning
- Machine Learning for Ocean Engineers
- Robotics Lab
- Mechatronic Systems
- Ship Maneuvering and Control

Core Research Competencies

- **Control Theory:** PD/PID Control Design, System Identification, State Estimation (Kalman Filters), Integral Line-of-Sight (I-LoS) Guidance.
- **Dynamical Systems:** Vessel Dynamic Modelling (MMG Models), Non-linear Dynamics, Stability Analysis.
- **Machine Learning:** Reinforcement Learning (RL), Neural Networks (MLP), Supervised Learning, Object Detection (YOLOv4).
- **Motion Planning:** Artificial Potential Fields (APFs), Rapidly-exploring Random Trees (RRT), Kino-dynamic Planning.

Programming & Software

- **Languages:** Python, MATLAB, C++, LaTeX, Mathematica
- **Frameworks & Simulators:** Robot Operating System (ROS), Simulink, Gazebo, Docker
- **Tools:** Git, Command Line Interface
- **Operating Systems:** Ubuntu Linux

AWARDS AND ACHIEVEMENTS

Virtual RobotX (VRX) ([Video summary](#) of the project)

(November 2021 – April 2022)

(A competition organized by RoboNation and Office of Naval Research)

- Placed **5th internationally** among **33 teams all over the globe** in Virtual RobotX Competition.
- Developed **autonomy** solutions for an unmanned surface vessel (USV) in **Gazebo** using **ROS**.
- Implemented **holonomic control** in three degrees of freedom using vectored thrust configuration on the USV.
- Implemented **LOS-Guidance**, Navigation and PID Control Algorithms for the USV to perform autonomous tasks.
- Applied **state-estimation techniques** like **Kalman Filter** using Robot Localization Package (ROS package) to fuse noisy GNSS and IMU sensor data, and thus creating a robust and reliable model for estimating the vessel's states.
- Trained a **YOLO v4** Deep Learning Model on a **custom dataset** to perform object localization.
- **Investigated LiDAR and Camera** data stream to localize obstacles to perform obstacle avoidance.

Oceans22 Student Hackathon

(February 2022)

(A student hackathon organized by the IEEE OCEANS-22 Conference Chennai)

- **Won the first prize** among 15 different teams at the OCEANS22 Student Hackathon in the Hardware Interface module.
- **Interfaced** the provided analog sensors with a single board computer to visualize real-time data

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

Teaching Assistant Ship Structures (OE3015) and Ship Maneuvering and Control (OE3036)

(July- Nov 2022)

(Instructor – Dr. Abhilash Sharma Somayajula)

- Facilitated effective student learning and doubt clarification under the guidance of the course instructor.
- Designed thought-provoking assignments to reinforce control theory and mechanics concepts and encourage analytical thinking.
- Organized and delivered multiple ROS tutorial sessions for nearly **50 undergraduate students** to extend their learning into robotics and software tools.

4 (TN) Air Sqn Tech NCC

(Aug 2018 - March 2020)

- **Volunteered** for the National Cadet Corps (NCC) program of the Institute.
- **Participated** in the 15th August Independence Day of India parade of the Institute.

Hobbies

- Passionate follower of **Formula 1 racing**, with a strong appreciation for strategy, performance engineering, and team dynamics in high-pressure environments. I enjoy watching races on weekend (especially street circuit) and occasionally analyze race telemetry data.
- I sometimes write blogs on Quora regarding cricket and general stuff and love playing Badminton.